A Unified Framework of Cloud Detection and Removal Based on Low-Rank and Group Sparse Regularizations for Multitemporal Multispectral Images

Teng-Yu Ji, Delin Chu, *Senior Member, IEEE*, Xi-Le Zhao, and Danfeng Hong, *Senior Member, IEEE*

***Abstract*— The existing cloud removal methods either need a cloud mask as prior knowledge or detect clouds before cloud removal processing, i.e., the detection and removal processes are separate. In this article, we propose a box-constrained (BC) smooth low-rank plus group sparse model to simultaneously detect and remove clouds, by formulating the degraded data as the summation of image and cloud components. For the cloud component, we propose a group sparse function along the spectral dimension. This is motivated by our observations that: 1) one tube is contaminated by clouds if any pixels of this tube are contaminated by clouds; 2) the positions of tubes, which are taken at different times, contaminated by clouds are different. For the image component, we propose to use a tensor rank based on the tensor singular value decomposition. The tensor rank characterizes the global property of the image component and could not keep the cloud-free information unchanged. To address the problem, we introduce a BC on the image component to force its cloud-free information to be equal to the corresponding values of observed data. Owing to the BC, the proposed model integrates the cloud detection and removal processes so that the two processes could promote each other and result in a promising result. To solve the proposed model, we develop an efficient algorithm that can generate the cloud mask, image component, and cloud component alternately. Extensive experiments on synthetic and real data show that the proposed method is competitive compared with the completion and other cloud removal methods.**

***Index Terms*— Box constraint (BC), cloud detection, cloud removal, group sparsity, low-rank tensor completion.**

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I. Introduction

THE observed optical remotely sensed data are generally covered by clouds, which leads to lose some important information and limits the subsequent applications, such as unmixing [1]–[4], classification [5]–[7], and fusion [8]–[12]. It is a challenging task to estimate the information covered by clouds or to characterize the cloud property, as the clouds have various shapes, heights, and distributions. According to the height of the cloud, the clouds removal processing could be divided into two cases, i.e., thin and thick cloud removal.

A. Related Works

According to whether the cloud mask is given, the cloud removal methods could be classified into two categories: methods with a given cloud mask, e.g., matrix/tensor completion methods and blind cloud removal methods. In the case of thin clouds, some blind methods, e.g., [13]–[17], and methods with given mask, e.g., [18]–[22], are proposed to estimate the missing information. Li *et al.* [13] propose to remove thin clouds based on dark channel prior, by observing that the thin clouds are semitransparent and are similar to fog. By considering the statistical characters of image information, Chun *et al.* [16] and Shen *et al.* [20] propose a homomorphic filter-based method to remove the thin clouds. Recently, Liu *et al.* [15] and Xu *et al.* [22] reconstruct the information covered by thin cloud based on the cloud physical transmission. With the development of deep learning, some deep methods have been designed for thin cloud removal, e.g., convolutional neural network (CNN)-based methods [18], [23], generative adversarial network (GAN)-based methods [14], [17], [21], and residual network-based methods [19].

As the thick clouds cause complete obstruction with loss of information, it is necessary to introduce the auxiliary images for reconstructing the contaminated information. In general, the data taken on the same area but at different times, which are termed as multitemporal data, are a good choice as the reference image to fill the missing information. For example, by borrowing the reference information from other temporal data, replace-based [24]–[26], inpainting [27]–[29], fusion [30], [31], and dictionary sparse [32]–[34] methods have been developed to

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remove the thick clouds and reconstruct the corresponding information. Besides these methods, the works [35] and [36] propose a B-spline model and an autoregression-based method to recover the cloud-contaminated area. Some cloud removal methods based on transmission principles are proposed to solve the problem [37], [38]. Recently, deep learning methods, e.g., GAN [39], CNN [40], [41], and deep prior [42], are developed to reconstruct the information covered by clouds for multitemporal remotely sensed images. In addition to referring to the multitemporal data, multisource data, e.g., synthetic aperture radar (SAR) data, are employed to remove the thick clouds. Gao *et al.* [43] reconstruct the information covered by thick clouds by fusing the optical and SAR images based on GAN. Li *et al.* [44] propose to estimate missing information based on a convolutional-mapping-deconvolutional network. Meraner *et al.* [45] propose to use a deep residual neural network to remove the cloud by fusing SAR and optical data. Ebel *et al.* [46] propose to remove cloud based on cycle-consistent GAN.

It is necessary to notice that the cloud mask should be given for the above methods for thick clouds removal. Therefore, these methods should just need focus on the missing areas and develop the relationships between the missing areas and the corresponding parts of other temporal data or of other source data, e.g., SAR data. These methods with a given cloud mask could reconstruct promising results. However, it is a difficult task to obtain an accurate cloud mask as the diversity of clouds. Thus, the results depend on the accuracy of the cloud mask, i.e., large or small mask both influences the estimated quality.

The second class of methods to remove the thick clouds is blind methods, where the cloud mask is not a necessary prior. Wen *et al.* [47] propose a two-pass method to remove thick clouds, where the first step is to detect the cloud mask with a plain robust principal component analysis (RPCA) method and the second step is to reconstruct the missing information with a discriminated RPCA. Zhang *et al.* [48] propose a coarse-to-fine method to remove clouds, where they characterize each cloud image with a group structured sparsity constraint and finally use a discriminative RPCA to improve the result. Chen *et al.* [49] propose a spatial–spectral total-variation regularized low-rank sparsity decomposition method. Subsequently, Duan *et al.* [50] propose a temporal smooth spatial–spectral total-variation regularized low-rank group sparsity decomposition method. These two works consider the low-rank property of underlying data by reshaping the data into a matrix, which may destroy the intrinsic high-dimensional structures and do not consider the correlation between bands although the sparsity of cloud with each band is taken into account.

B. Motivation

The existing blind methods could be improved further. First, these methods separate the removal method into two parts [47], [48], i.e., detection and reconstruction, or three parts [49], [50], i.e., detection, reconstruction, and correction. This separation leads these separate parts to do their jobs independently and ignore the complementary relationships of these parts. Moreover, it is not sufficient to characterize the cloud property by only considering each band, since there are some correlations between the clouds in each band of mulitspectral data.

To better characterize the cloud features, we propose a group sparse function along the spectral dimension. The inspiration comes from three aspects: 1) the positions of clouds for all bands at one certain time are almost the same because the power of waves of multispectral sensors breaks through the clouds are similar; 2) part of pixels are contaminated by clouds at one certain time; and 3) the data taken at different times are contaminated by different clouds, i.e., the type and position are different, since the clouds vary with time. We show the data taken over France on February 24, 2020 (top two rows in Fig. 1) and October 3, 2019 (bottom two rows in Fig. 1) as an example. From the second and fourth rows, we can observe that the type and position of clouds for the data taken at the same time are almost the same and that the data contains cloud-free and cloud-contaminated pixels simultaneously. By comparing the top two and bottom two rows, we could know that the type and position of clouds are different when the taken time is different. These observations yield that the tubes are nonzeros or vectors of zeros, the multitemporal data contains sparse nonzero tubes. As a remedy, we could employ the group sparse function along the spectral dimension to regularize the cloud component.

For the image component, the rank function is a powerful tool to characterize its global correlation, which has been used in many multispectral image processing problems [51]–[56]. In this article, we propose to use the tensor rank based on tensor singular value decomposition to characterize the global correlation. However, the tensor rank takes into consideration the relationships of all pixels between different temporal modes, rather than the relationship of missing areas. That means the low-rank-based methods would change the cloud-free information and reduce the recovery quality if there is no correction strategy [49], [50]. To tackle the problem, the pixels that are not contaminated by clouds should be detected and retained unchanged just like the completion methods [29], [57]–[65]. Fortunately, the group sparse function along the spectral dimension could characterize the cloud property well and output an accurate cloud component. Therefore, a cloud mask could be generated and used to guarantee the cloud-free information of the image component is unchanged. Based on the above-mentioned observations, we propose a box constraint (BC) to force the entries, which are not contaminated by clouds, to be equal to the corresponding entries of the observed data.

Combining the above-mentioned analysis and the property that are similar to the adjacent temporal data, we propose a BC smooth low-rank plus group sparse (BC-SLRpGS) model for cloud detection and removal. Fig. 2 shows the framework of the proposed method. To solve the resulting proposed model, we develop an efficient algorithm, in which the cloud mask is first obtained via a hard thresholding function and then the image and cloud components are obtained by the proximal alternating minimization (PAM)-based algorithm [66].

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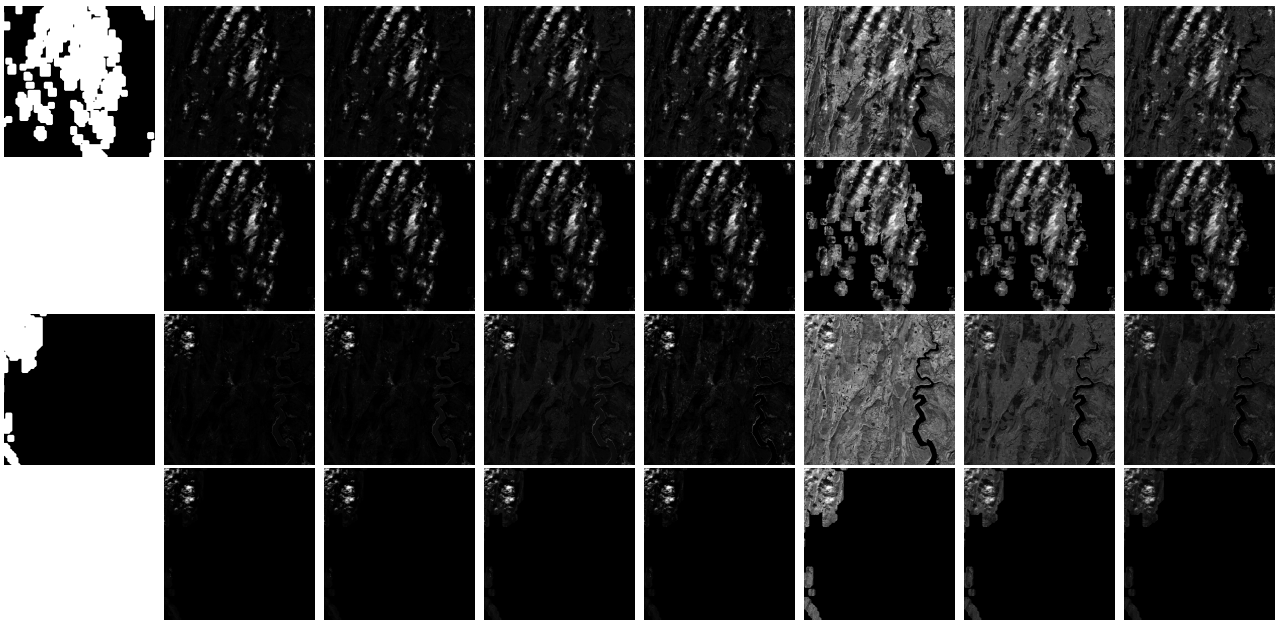

Fig. 1. France data taken on February 24, 2020 (top two rows) and October 3, 2019 (bottom two rows), the corresponding pesudocolor images are shown in Fig. 9. The first and third rows are the observed data with seven bands, and the first image is the cloud mask for each temporal data. The second and fourth rows are the cloud component for each temporal data.

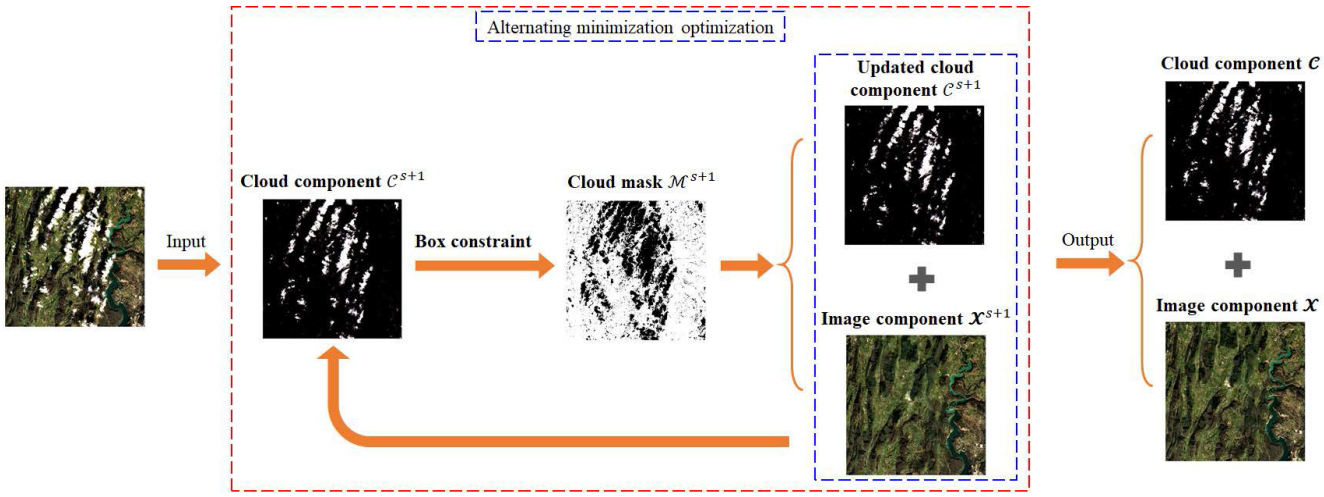

Fig. 2. Flowchart of the proposed method. The red dash-rectangle displays the iterative process of the proposed method for cloud detection and removal.

Extensive experiments on synthetic and real data show that the proposed method is competitive compared with the tensor completion and other cloud removal methods.

The contributions of the proposed method are as follows.

1) We propose a group sparse function along a spectral dimension, which takes advantage of the correlation of clouds between bands and could characterize the cloud's property more accurately compared with sparse function.
2) With the proposed group sparse function, we propose a smooth low-rank plus group sparse model that integrates cloud detection and removal processes by introducing a BC, which enforces the cloud-free areas of image component in each iteration are the same as that of observed data.
3) An efficient algorithm is developed to solve the resulting model. The cloud mask is obtained by a hard thresholding function, and the image and cloud components are obtained by the PAM-based algorithm. Extensive experiments on synthetic and real data show that the proposed method is competitive compared with the tensor completion and other cloud removal methods.

C. Notations

In this article, we denote scalars by lower case letters, e.g., x, vectors by bold lower case letters, e.g., $\boldsymbol{x}$, matrices by bold upper case letters, e.g., $\boldsymbol{X}$, and tensors by bold calligraphic letters, e.g., $\boldsymbol{\mathcal{X}}$. The multitemporal remotely sensed image is a fourth-order tensor and is denoted as $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{m\times n\times p\times q}$, where m, n, p, and q denote the height-spatial, width-spatial, spectral, and temporal dimensions, respectively. The mode-r ***fibers*** are vectors by fixing all mode indexes except r,

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e.g., $\boldsymbol{\mathcal{X}}_{:,j,k,l}$, $\boldsymbol{\mathcal{X}}_{i,:,k,l}$, $\boldsymbol{\mathcal{X}}_{i,j,:,l}$, and $\boldsymbol{\mathcal{X}}_{i,j,k,:}$ are modes 1, 2, 3, and 4 fibers, respectively. Especially, the mode-3 fibers are also called as ***tubes***. The mode-r ***unfolding*** is denoted by $\boldsymbol{X}_{(r)} = \text{Unfold}_r(\boldsymbol{\mathcal{X}})$, whose columns are the mode-r fibers. The corresponding inverse operator is defined as $\boldsymbol{\mathcal{X}} = \text{Fold}_r(\boldsymbol{X}_{(r)})$. The mode-$r$ tensor-matrix product between a tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and a matrix $\boldsymbol{Z} \in \mathbb{R}^{m \times I_r}$ is defined as $\boldsymbol{\mathcal{Y}} = \boldsymbol{\mathcal{X}} \times_r \boldsymbol{Z} = \text{Fold}_r(\boldsymbol{Z}\boldsymbol{X}_{(r)}) \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_{r-1} \times m \times I_{r+1} \times \cdots \times I_N}$. The Frobenius norm of tensor $\boldsymbol{\mathcal{X}}$ is defined by $\|\boldsymbol{\mathcal{X}}\|_F := (\sum_{i,j,k,l} |x_{i,j,k,l}|^2)^{1/2}$, where $x_{i,j,k,l}$ is the (i, j, k, l)th element of $\boldsymbol{\mathcal{X}}$.

The rest of this article is organized as follows. Section II presents the proposed cloud removal model and the optimization algorithm. Section III studies the performance of the proposed method with synthetic and real-multitemporal multispectral data. Section IV includes some conclusions.

II. Methodology

Suppose $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{m \times n \times p \times q}$ is the observed data containing clouds, the degraded processing can be formulated as

$$\boldsymbol{\mathcal{Y}} = \boldsymbol{\mathcal{X}} + \boldsymbol{\mathcal{C}} + \boldsymbol{\mathcal{N}} \tag{1}$$

where $\boldsymbol{\mathcal{X}}$ is the desired image component, $\boldsymbol{\mathcal{C}}$ is the cloud component whose nonzero entries denote the cloud, and $\boldsymbol{\mathcal{N}}$ is the model noise. The goal of our work is to estimate both image and cloud components from the observed data.

A. Cloud and Image Characteristic Analysis

To accurately formulate the cloud component $\boldsymbol{\mathcal{C}}$, we find that the cloud-contained remotely sensed images have three properties. The first is that all bands of one area are contaminated by almost the same clouds if any band of this area is cloud-contained. The second is that there are cloud-free and cloud-contained pixels simultaneously in the whole field of view image. The third is that cloud position varies with time; therefore, we assume that one certain location is contaminated by cloud in one temporal mode, while the pixels of this location at other temporal modes are cloud-free. Inspired by the above-mentioned second and third properties, the cloud component $\boldsymbol{\mathcal{C}}$ could be characterized by a sparse function, e.g., l_1-norm used in [49].

However, l_1-norm characterizes the global sparse property of cloud components and ignores the correlations between the bands. To address this problem, let us further study the above-mentioned three properties. Suppose the (i, j, k, l)th pixel of cloud component $\boldsymbol{\mathcal{C}} \in \mathbb{R}^{m \times n \times p \times q}$ is contaminated by cloud, i.e., $\boldsymbol{\mathcal{C}}_{i,j,k,l} \neq 0$. Considering the (i, j)th location.

1) if the location is contaminated by clouds at band k_b and time l_t (i.e., $\boldsymbol{\mathcal{C}}_{i,j,k_b,l_t} \neq 0$). a) According to the first property, the value of tube $\boldsymbol{\mathcal{Y}}_{i,j,:,l_t}$ denotes the clouds, namely, the pixels $\boldsymbol{\mathcal{C}}_{i,j,k,l_t} \neq 0$ for $k = 1, 2, \ldots, p$. b) According to the third property, the tubes $\boldsymbol{\mathcal{C}}_{i,j,:,l} = 0$ for $1 \leq l \leq q$ and $l \neq l_t$.

2) According to the second property, there exists $(i', j', :, l_t)$ satisfying $\boldsymbol{\mathcal{C}}_{i',j',:,l_t} = 0$, namely, the (i', j')th location at time l_t is cloud free.

Therefore, we conclude that the cloud component $\boldsymbol{\mathcal{C}}$ can be characterized by a group sparse function along the spectral dimension. Specifically, the tubes of $\boldsymbol{\mathcal{C}}$ are nonzeros if the pixel is contaminated by clouds, or zeros, otherwise. Mathematically, the group sparse function of $\boldsymbol{\mathcal{C}}$ along the spectral dimension is defined by

$$\begin{aligned} \mathcal{S}(\boldsymbol{\mathcal{C}}) &= \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{l=1}^{q} \left\|\boldsymbol{\mathcal{C}}_{i,j,:,l}\right\|_2 \\ &= \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{l=1}^{q}\left(\sum_{k=1}^{p}\boldsymbol{\mathcal{C}}_{i,j,k,l}^2\right)^{1/2} \end{aligned} \tag{2}$$

where $\boldsymbol{\mathcal{C}}_{i,j,:,l}$ denotes the tubes of $\boldsymbol{\mathcal{C}}$.

The image component $\boldsymbol{\mathcal{X}}$ is a fourth-order tensor, whose global correlation can be characterized by the tensor rank function [57], [58], [67]–[70], e.g., CP-rank, Tucker-rank, tubal-rank, multirank, and so on. In this work, we adopt the multirank as the regularization function of $\boldsymbol{\mathcal{X}}$. Since the multirank is defined for the third tensor, we define the regularization function as follows:

$$\text{rank}_m(\boldsymbol{\mathcal{X}}) = \sum_{r=1}^{pq}\|\boldsymbol{Z}_r\|_* \tag{3}$$

where $\boldsymbol{Z}_r$ is the rth frontal slices of $\boldsymbol{\mathcal{Z}} = \mathcal{R}(\boldsymbol{\mathcal{X}}) \times_3 \boldsymbol{Q}^T$; here, $\boldsymbol{Q} \in \mathbb{R}^{pq \times pq}$ is an orthogonal matrix and $\mathcal{R}(\cdot) : \mathbb{R}^{m \times n \times p \times q} \to \mathbb{R}^{m \times n \times pq}$ is a reshape operator that maps the fourth-order tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{m \times n \times p \times q}$ into a third-order tensor $\mathcal{R}(\boldsymbol{\mathcal{X}}) \in \mathbb{R}^{m \times n \times pq}$.

Besides, we observe that the adjacent temporal data is similar, which can be characterized by the following function:

$$\|\mathcal{D}(\boldsymbol{\mathcal{X}})\|_F^2 = \left\|\boldsymbol{D}\boldsymbol{X}_{(4)}\right\|_F^2 \tag{4}$$

where $\mathcal{D}(\cdot)$ denotes the difference operator along the fourth mode, $\boldsymbol{X}_{(4)}$ is the mode-4 unfolding of $\boldsymbol{\mathcal{X}}$, and $\boldsymbol{D} \in \mathbb{R}^{q \times q}$ is a difference matrix whose (i, j)th entries are defined as follows:

$$\boldsymbol{d}_{i,j} = \begin{cases} 1, & \text{if } i = j \\ -1, & \text{if } j = i + 1 \\ -1, & \text{if } i = q \text{ and } j = 1 \\ 0, & \text{otherwise.} \end{cases}$$

As a conclusion, the general cloud removal model can be formulated as

$$\begin{aligned} &\min_{\boldsymbol{\mathcal{X}},\boldsymbol{\mathcal{C}}} \text{rank}_m(\boldsymbol{\mathcal{X}}) + \lambda\mathcal{S}(\boldsymbol{\mathcal{C}}) + \frac{\tau}{2}\|\mathcal{D}(\boldsymbol{\mathcal{X}})\|_F^2 \\ &\text{s.t. } \boldsymbol{\mathcal{Y}} = \boldsymbol{\mathcal{X}} + \boldsymbol{\mathcal{C}} \end{aligned} \tag{5}$$

where $\lambda > 0$ and $\tau > 0$ are the regularization parameters.

B. Proposed Model

The general model (5) considers the global properties of image and cloud components. It should be noted that the cloud-free areas of $\boldsymbol{\mathcal{X}}$ obtained from the model may be changed compared with those of observed data, which reduces the reconstruction quality. To address this issue, Shen *et al.* [31] and Li *et al.* [37] propose to introduce a residual/error correction strategy after obtaining the image component. Another method is to replace the cloud-free parts of $\boldsymbol{\mathcal{X}}$ with the corresponding parts of observed data after

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finishing the iteration and obtain $\mathcal{X}$ and $\mathcal{C}$. For example, Chen *et al.* [49] do a cloud/shadow detection and information compensation processing after obtaining the desired data $\mathcal{X}$. These methods are composed of some independent steps, i.e., detection, reconstruction, and correction, and ignore the interrelation of these signs of progress. To tackle this problem, we integrate the cloud detection and removal processing into one model and update the cloud-free areas of the image component with the corresponding areas of observed data. Specifically, we generate a cloud mask tensor $\mathcal{M} \in \mathbb{R}^{m\times n\times p\times q}$ based on the cloud component $\mathcal{C}$ by defining a function as $\mathcal{M} = f(\mathcal{C})$, whose tubes are

$$\mathcal{M}_{i,j,:,l} = \begin{cases} 1, & \text{if } \dfrac{1}{p}\displaystyle\sum_{k=1}^{p} |c_{i,j,k,l}| < \epsilon \\ 0, & \text{else} \end{cases} \tag{6}$$

where $\epsilon \geq 0$ is a given thresholding value. The pixel $\mathcal{M}_{i,j,k,l} = 1$ denotes that the (i, j, k, l)th pixel of observed data $\mathcal{Y}$ is cloud-free, while $\mathcal{M}_{i,j,k,l} = 0$ denotes that the pixel is covered by cloud. Consequently, we proposed a BC-SLRpGS model, that is,

$$\begin{aligned} \min_{\substack{\mathcal{X}\in\mathbb{S},\mathcal{C},\mathcal{Z},\\ \mathcal{Q},\mathcal{M}}} & \sum_{r=1}^{pq}\|\boldsymbol{Z}_r\|_* + \lambda\sum_{i,j,l}\left\|\mathcal{C}_{i,j,:,l}\right\|_2 + \frac{\tau}{2}\left\|\boldsymbol{D}\boldsymbol{X}_{(4)}\right\|_F^2 \\ \text{s.t. } & \mathcal{R}(\mathcal{X}) = \mathcal{Z}\times_3 \mathcal{Q}, \quad \mathcal{Q}^T\mathcal{Q} = \boldsymbol{I}, \\ & \mathcal{Y} = \mathcal{X} + \mathcal{C}, \ \ \mathcal{M} = f(\mathcal{C}) \end{aligned} \tag{7}$$

where λ and τ are two positive regularization parameters, $\boldsymbol{I} \in \mathbb{R}^{pq\times pq}$ is an identity matrix, and $\mathcal{X} \in \mathbb{S}$ is the proposed box constraint, here

$$\mathbb{S} = \{\mathcal{X} | \mathcal{M} \odot (\mathcal{Y} - \mathcal{X}) = 0\}$$

where $\odot$ denotes the Hadamard product.

C. Optimization Algorithm

In this section, we develop an optimization algorithm to solve the proposed model (7). Since the variables $(\mathcal{X}, \mathcal{C}, \mathcal{Z}, \mathcal{Q}, \mathcal{M})$ are coupled, we introduce an auxiliary variable $\mathcal{W} = \mathcal{Z}$ and rewrite the model (7) as

$$\begin{aligned} \min_{\substack{\mathcal{X}\in\mathbb{S},\mathcal{C},\mathcal{Z},\\ \mathcal{W},\mathcal{Q},\mathcal{M}}} & \sum_{r=1}^{pq}\|\boldsymbol{W}_r\|_* + \lambda\sum_{i,j,l}\left\|\mathcal{C}_{i,j,:,l}\right\|_2 + \frac{\tau}{2}\left\|\boldsymbol{D}\boldsymbol{X}_{(4)}\right\|_F^2 \\ \text{s.t. } & \mathcal{R}(\mathcal{X}) = \mathcal{Z}\times_3 \mathcal{Q}, \quad \mathcal{W} = \mathcal{Z}, \ \ \mathcal{Q}^T\mathcal{Q} = \boldsymbol{I} \\ & \mathcal{Y} = \mathcal{X} + \mathcal{C}, \quad \mathcal{M} = f(\mathcal{C}) \end{aligned} \tag{8}$$

where $\boldsymbol{W}_r$ is the rth frontal slice of $\mathcal{W}$.

The above-mentioned constrained problem can be solved via an alternatively minimizing strategy as follows:

$$\begin{cases} \mathcal{M}^{s+1} = f(\mathcal{C}^s) \\ (\mathcal{X}^{s+1}, \mathcal{C}^{s+1}, \mathcal{Z}^{s+1}, \mathcal{W}^{s+1}, \mathcal{Q}^{s+1}) \\ = \underset{\substack{\mathcal{X}\in\mathbb{S},\mathcal{C},\mathcal{Z},\\ \mathcal{W},\mathcal{Q}^T\mathcal{Q}=\boldsymbol{I}}}{\arg\min}\, h(\mathcal{X}, \mathcal{C}, \mathcal{Z}, \mathcal{W}, \mathcal{Q}) \end{cases} \tag{9}$$

where the first equation is to estimate the cloud mask according to the cloud component, and

$$\begin{aligned} & h(\mathcal{X}, \mathcal{C}, \mathcal{Z}, \mathcal{W}, \mathcal{Q}) \\ & = \sum_{r=1}^{pq}\|\boldsymbol{W}_r\|_* + \lambda\sum_{i,j,l}\left\|\mathcal{C}_{i,j,:,l}\right\|_2 \\ & \quad + \frac{\tau}{2}\left\|\boldsymbol{D}\boldsymbol{X}_{(4)}\right\|_F^2 + \frac{\beta_1}{2}\|\mathcal{R}(\mathcal{X}) - \mathcal{Z}\times_3 \mathcal{Q}\|_F^2 \\ & \quad + \frac{\beta_2}{2}\|\mathcal{W} - \mathcal{Z}\|_F^2 + \frac{\beta_3}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{C}\|_F^2 \end{aligned} \tag{10}$$

where $\beta_i > 0$ $(i = 1, 2, 3)$ are penalty parameters.

By employing the half quadratic splitting tips [71], the variables $(\mathcal{X}, \mathcal{C}, \mathcal{Z}, \mathcal{W}, \mathcal{Q})$ can be updated alternatively as follows:

$$\begin{cases} \mathcal{X}^{s+1} = \underset{\mathcal{X}\in\mathbb{S}}{\arg\min}\, h(\mathcal{X}, \mathcal{C}^s, \mathcal{Z}^s, \mathcal{W}^s, \mathcal{Q}^s) + \dfrac{\rho}{2}\left\|\mathcal{X} - \mathcal{X}^s\right\|_F^2 \\ \mathcal{Z}^{s+1} = \underset{\mathcal{Z}}{\arg\min}\, h(\mathcal{X}^{s+1}, \mathcal{C}^s, \mathcal{Z}, \mathcal{W}^s, \mathcal{Q}^s) + \dfrac{\rho}{2}\left\|\mathcal{Z} - \mathcal{Z}^s\right\|_F^2 \\ \mathcal{C}^{s+1} = \underset{\mathcal{C}}{\arg\min}\, h(\mathcal{X}^{s+1}, \mathcal{C}, \mathcal{Z}^{s+1}, \mathcal{W}^s, \mathcal{Q}^s) + \dfrac{\rho}{2}\left\|\mathcal{C} - \mathcal{C}^s\right\|_F^2 \\ \mathcal{W}^{s+1} = \underset{\mathcal{W}}{\arg\min}\, h(\mathcal{X}^{s+1}, \mathcal{C}^{s+1}, \mathcal{Z}^{s+1}, \mathcal{W}, \mathcal{Q}^s) \\ \qquad + \dfrac{\rho}{2}\left\|\mathcal{W} - \mathcal{W}^s\right\|_F^2 \\ \mathcal{Q}^{s+1} = \underset{\mathcal{Q}^T\mathcal{Q}=\boldsymbol{I}}{\arg\min}\, h(\mathcal{X}^{s+1}, \mathcal{C}^{s+1}, \mathcal{Z}^{s+1}, \mathcal{W}^{s+1}, \mathcal{Q}) \\ \qquad + \dfrac{\rho}{2}\left\|\mathcal{Q} - \mathcal{Q}^s\right\|_F^2 \end{cases} \tag{11}$$

where $\rho > 0$ is the proximal parameter and the superscript ^{s} denotes the corresponding updated result in the sth iteration.

1) *Updating $\mathcal{X}$:* $\mathcal{X}$-subproblem is to solve the following optimization problem:

$$\begin{aligned} \min_{\mathcal{X}\in\mathbb{S}} & \frac{\beta_1}{2}\left\|\mathcal{R}(\mathcal{X}) - \mathcal{Z}^s\times_3 \mathcal{Q}^s\right\|_F^2 + \frac{\tau}{2}\left\|\boldsymbol{D}\boldsymbol{X}_{(4)}\right\|_F^2 \\ & + \frac{\beta_3}{2}\left\|\mathcal{Y} - \mathcal{X} - \mathcal{C}^s\right\|_F^2 + \frac{\rho}{2}\left\|\mathcal{X} - \mathcal{X}^s\right\|_F^2. \end{aligned} \tag{12}$$

As $\mathcal{R}(\cdot)$ is a reversible reshape operator, the above-mentioned equation can be reformulated as the following:

$$\begin{aligned} \min_{\boldsymbol{X}} & \frac{\beta_1}{2}\left\|\boldsymbol{X}_{(4)} - \mathcal{R}^{-1}(\mathcal{Z}^s\times_3 \mathcal{Q}^s)_{(4)}\right\|_F^2 + \frac{\tau}{2}\left\|\boldsymbol{D}\boldsymbol{X}_{(4)}\right\|_F^2 \\ & + \frac{\beta_3}{2}\left\|\boldsymbol{Y}_{(4)} - \boldsymbol{X}_{(4)} - \boldsymbol{C}_{(4)}^s\right\|_F^2 + \frac{\rho}{2}\left\|\boldsymbol{X}_{(4)} - \boldsymbol{X}_{(4)}^s\right\|_F^2 \end{aligned} \tag{13}$$

where $\mathcal{R}^{-1}(\cdot) : \mathbb{R}^{m\times n\times pq} \to \mathbb{R}^{m\times n\times p\times q}$ denotes the inverse operator of $\mathcal{R}(\cdot)$ and reshapes the third tensor into the original fourth format. Obviously, $\mathcal{X}$ has the following closed-form solution:

$$\begin{aligned} & \boldsymbol{X}_{(4)}^{s+1} \\ & = \left((\beta_1 + \beta_3 + \rho)\boldsymbol{I} + \tau\boldsymbol{D}^T\boldsymbol{D}\right)^{-1} \\ & \quad \times\left(\beta_1\mathcal{R}^{-1}\left(\mathcal{Z}^s\times_3 \mathcal{Q}^s\right)_{(4)} + \beta_3\left(\boldsymbol{Y}_{(4)} - \boldsymbol{C}_{(4)}^s\right) + \rho\boldsymbol{X}_{(4)}^s\right) \end{aligned} \tag{14}$$

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where $\boldsymbol{I} \in \mathbb{R}^{q\times q}$ is an identity matrix. Combing the BC, the image component is obtained as follows:

$$\boldsymbol{\mathcal{X}}^{s+1} = (1-\boldsymbol{\mathcal{M}}^{s+1}) \odot \text{Fold}_4\left(\boldsymbol{X}_{(4)}^{s+1}\right) + \boldsymbol{\mathcal{M}}^{s+1} \odot \boldsymbol{\mathcal{Y}} \tag{15}$$

where $\text{Fold}_{(4)}(\cdot)$ is the inverse operator of unfolding along with the fourth mode, i.e., the operator reshapes the matrix into a tensor along with the fourth mode.

2) Updating $\boldsymbol{\mathcal{Z}}$: We present the $\boldsymbol{\mathcal{Z}}$-subproblem, that is,

$$\min_{\boldsymbol{\mathcal{Z}}} \frac{\beta_1}{2}\left\|\mathcal{R}(\boldsymbol{\mathcal{X}}^{s+1}) - \boldsymbol{\mathcal{Z}} \times_3 \boldsymbol{Q}^s\right\|_F^2 + \frac{\beta_2}{2}\left\|\boldsymbol{\mathcal{W}}^s - \boldsymbol{\mathcal{Z}}\right\|_F^2 + \frac{\rho}{2}\left\|\boldsymbol{\mathcal{Z}} - \boldsymbol{\mathcal{Z}}^s\right\|_F^2. \tag{16}$$

Therefore, the closed-form solution can be obtained

$$\boldsymbol{\mathcal{Z}}^{s+1} = \frac{\beta_1\mathcal{R}(\boldsymbol{\mathcal{X}}^{s+1}) \times_3 (\boldsymbol{Q}^s)^T + \beta_2\boldsymbol{\mathcal{W}}^s + \rho\boldsymbol{\mathcal{Z}}^s}{\beta_1+\beta_2+\rho}. \tag{17}$$

3) Updating $\boldsymbol{\mathcal{C}}$: The corresponding subproblem is as follows:

$$\min_{\boldsymbol{\mathcal{C}}} \sum_{i,j,l} \lambda\left\|\boldsymbol{\mathcal{C}}_{i,j,:,l}\right\|_2 + \frac{\rho}{2}\left\|\boldsymbol{\mathcal{C}} - \boldsymbol{\mathcal{C}}^s\right\|_F^2 + \frac{\beta_3}{2}\left\|\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{X}}^{s+1} - \boldsymbol{\mathcal{C}}\right\|_F^2. \tag{18}$$

The problem can be solved by splitting it into the following subproblems:

$$\min_{\boldsymbol{\mathcal{C}}} \sum_{i,j,l} \left\{\lambda\left\|\boldsymbol{\mathcal{C}}_{i,j,:,l}\right\|_2 + \frac{\rho}{2}\left\|\boldsymbol{\mathcal{C}}_{i,j,:,l} - \boldsymbol{\mathcal{C}}^s_{i,j,:,l}\right\|_F^2 + \frac{\beta_3}{2}\left\|\boldsymbol{\mathcal{C}}_{i,j,:,l} - \left(\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{X}}^{s+1}\right)_{i,j,:,l}\right\|_F^2\right\}. \tag{19}$$

By incorporating the last two terms, the above problem is equivalent to the following:

$$\min_{\boldsymbol{\mathcal{C}}} \sum_{i,j,l} \left\{\lambda\left\|\boldsymbol{\mathcal{C}}_{i,j,:,l}\right\|_2 + \frac{\beta_3+\rho}{2}\left\|\boldsymbol{\mathcal{C}}_{i,j,:,l} - \boldsymbol{\mathcal{P}}_{i,j,:,l}\right\|_F^2\right\} \tag{20}$$

where $\boldsymbol{\mathcal{P}}_{i,j,:,l}$ is the tube of $\boldsymbol{\mathcal{P}} = (\beta_3(\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{X}}^{s+1}) + \rho\boldsymbol{\mathcal{C}}^s/\beta_3 + \rho)$. Therefore, the solution of tube $\boldsymbol{\mathcal{C}}_{i,j,:,l}$ is obtained as follows:

$$\boldsymbol{\mathcal{C}}^{s+1}_{i,j,:,l} = \max\left(\left\|\boldsymbol{\mathcal{P}}_{i,j,:,l}\right\|_2 - \frac{\lambda}{\beta_3+\rho}, 0\right)\frac{\boldsymbol{\mathcal{P}}_{i,j,:,l}}{\left\|\boldsymbol{\mathcal{P}}_{i,j,:,l}\right\|_2} \tag{21}$$

where we denote $(0/0) = 1$.

4) Updating $\boldsymbol{\mathcal{W}}$: $\boldsymbol{\mathcal{W}}^{s+1}$ is solved by the following problem:

$$\min_{\boldsymbol{\mathcal{W}}} \sum_{r=1}^{pq} \|\boldsymbol{W}_r\|_* + \frac{\beta_2}{2}\left\|\boldsymbol{\mathcal{W}} - \boldsymbol{\mathcal{Z}}^{s+1}\right\|_F^2 + \frac{\rho}{2}\left\|\boldsymbol{\mathcal{W}} - \boldsymbol{\mathcal{W}}^s\right\|_F^2. \tag{22}$$

By incorporating the last two terms, the above-mentioned problem is equivalent to the following:

$$\min_{\boldsymbol{\mathcal{W}}} \sum_{r=1}^{pq} \left\{\|\boldsymbol{W}_r\|_* + \frac{\beta_2+\rho}{2}\|\boldsymbol{W}_r - \boldsymbol{R}_r\|_F^2\right\} \tag{23}$$

where $\mathcal{R} = (\beta_2\boldsymbol{\mathcal{Z}}^{s+1} + \rho\boldsymbol{\mathcal{W}}^s/\beta_2 + \rho)$ and $\boldsymbol{R}_r$ denotes the rth frontal slice of $\mathcal{R}$. Therefore, $\boldsymbol{\mathcal{W}}$ can be solved slice by slice (frontal slices). For a given r, the solution of $\boldsymbol{W}_r$ can be obtained by a singular value thresholding (SVT) operator, that is,

$$\boldsymbol{W}_r^{s+1} = \text{SVT}\left(\boldsymbol{R}_r, \frac{1}{\beta_2+\rho}\right) = \boldsymbol{U}\mathcal{T}_{\frac{1}{\beta_2+\rho}}(\boldsymbol{\Sigma})\boldsymbol{V}^T \tag{24}$$

where, $\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \text{SVD}(\boldsymbol{R}_r)$, here $\text{SVD}(\cdot)$ denotes the singular value decomposition (SVD), and $\mathcal{T}_{(1/\beta_2+\rho)}(\boldsymbol{\Sigma})$ is a diagonal matrix whose jth diagonal entry is $\max(\sigma_j - (1/\beta_2+\rho), 0)$, where σ_j is the jth diagonal value of $\boldsymbol{\Sigma}$.

5) Updating $\boldsymbol{Q}$: The corresponding optimization problem is as follows:

$$\min_{\boldsymbol{Q}^T\boldsymbol{Q}=\boldsymbol{I}} \frac{\beta_1}{2}\left\|\mathcal{R}(\boldsymbol{\mathcal{X}}^{s+1}) - \boldsymbol{\mathcal{Z}}^{s+1} \times_3 \boldsymbol{Q}\right\|_F^2 + \frac{\rho}{2}\left\|\boldsymbol{Q} - \boldsymbol{Q}^s\right\|_F^2. \tag{25}$$

Since the subproblem has an orthogonal constraint, we rewrite it as the following equivalent problem:

$$\max_{\boldsymbol{Q}^T\boldsymbol{Q}=\boldsymbol{I}} \text{tr}\left(\beta_1\left(\mathcal{R}\left(\boldsymbol{\mathcal{X}}^{s+1}\right)\right)_{(3)}\left(\boldsymbol{Z}_{(3)}^{s+1}\right)^T + \rho\boldsymbol{Q}^s\right)\boldsymbol{Q}^T \tag{26}$$

where $\text{tr}(\cdot)$ denotes the matrix trace. Let $\hat{\boldsymbol{U}}\hat{\boldsymbol{\Sigma}}\hat{\boldsymbol{V}}^T = \text{SVD}(\beta_1(\mathcal{R}(\boldsymbol{\mathcal{X}}^{s+1}))_{(3)}(\boldsymbol{Z}_{(3)}^{s+1})^T + \rho\boldsymbol{Q}^s)$, then the closed-form solution of $\boldsymbol{Q}$-subproblem is

$$\boldsymbol{Q}^{s+1} = \hat{\boldsymbol{U}}\hat{\boldsymbol{V}}^T. \tag{27}$$

An analysis earlier is summarized as Algorithm 1. The initialization of image and cloud components are zero tensors. If the BC is processed from the first iteration, the cloud masks $\boldsymbol{\mathcal{M}}$ would always keep being the tensor of zeros. That means the BC loses its effectiveness. To obtain a promising cloud mask, we set that the BC is activated if the error of cloud component satisfies $\varepsilon_{\boldsymbol{\mathcal{C}}} < 0.005$. At each iteration, the cost of computing $\boldsymbol{\mathcal{M}}$ is $O(mnpq)$, of computing $\boldsymbol{\mathcal{X}}$ is $O(mnp^2q^2 + mnpq^2 + 2q^3)$, of computing $\boldsymbol{\mathcal{Z}}$ is $O(mnp^2q^2)$, of computing $\boldsymbol{\mathcal{C}}$ is $O(mnpq)$, of computing $\boldsymbol{\mathcal{W}}$ is $O(2mn\min\{m,n\}pq + m\min\{m,n\}^2pq)$, and of computing $\boldsymbol{Q}$ is $O(mnp^2q^2 + 2p^3q^3)$. In summary, the computational cost at each iteration is $O((3mn+2pq)p^2q^2)$.

III. Experiments

In this section, we test the performance of proposed method with synthetic and real multitemporal multispectral remotely sensed images compared with high accuracy low rank tensor completion (HaLRTC) based on Tucker decomposition [67], tensor singular value decomposition method (t-SVD) [68], tensor completion by parallel matrix factorization based on tensor train decomposition (TMacTT) [69], tensor ring low-rank factors (TRLRF) [70], augmented Lagrangian method with inexact proximal gradients (ALM-IPG) [29], and total-variation regularized low-rank sparsity decomposition with information compensation method (TVLRSDC) [49]. It is noted that HaLRTC, t-SVD, TMacTT, TRLRF, and ALM-IPG are low-rank completion-based methods, i.e., it is necessary for these methods to introduce cloud masks as prior information. For ALM-IPG, it adopts temporal smooth prior and

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TABLE I
COMPARISONS OF ALL COMPARED METHODS. THE "YES" MEANS THAT THE PRIOR IS ADOPTED IN THE CORRESPONDING METHOD, WHILE "NO" MEANS THAT THE PRIOR IS NOT USED

Method	Mask Prior	Low-rank Prior	Smooth Prior	Sparse Prior	Box Constraint
HaLRTC	Yes	Tucker rank	No	No	No
t-SVD	Yes	multi rank	No	No	No
TMacTT	Yes	TT rank	No	No	No
TRLRF	Yes	TR rank	No	No	No
ALM-IPG	Yes	matrix rank	Temporal smooth	No	No
TVLRSDC	No	matrix rank	Spatial and spectral smooth	Sparse of each band	No
BC-SLRpGS	No	multi rank	Temporal smooth	Group sparse along spectral dimension	Yes

Algorithm 1 BC-SLRpGS Cloud Detection and Removal Algorithm

Input: The observed cloud-contained data $\mathcal{Y}$ and parameters λ, $\beta_1 = \beta_2 = \beta_3$, and τ.

Initialize: $\mathcal{X}^0 = \mathbf{0}$, $\mathcal{C}^0 = \mathbf{0}$, $\mathcal{W}^0 = \mathbf{0}$, $\rho = 0.01$, tolerance for stopping criterion $tol = 10^{-5}$, and $S = 2000$.
for $s = 1$ to S **do**
  Update $\mathcal{M}$ by (6);
  Update $\mathcal{X}$ by (15);
  Set $\mathcal{M} \odot \mathcal{C} := 0$;
  Update $\mathcal{Z}$ by (17);
  Update $\mathcal{C}_{i,j,:,l}$ by (21) for each mode-3 tube;
  for $i = 1$ to pq **do**
    Update $\boldsymbol{W}_r$ by (24);
  end for
  Update $\boldsymbol{Q}$ by (27);
  Update error $\varepsilon_{\mathcal{X}} = \left\|\mathcal{X}^{s+1} - \mathcal{X}^s\right\|_F / \left\|\mathcal{X}^s\right\|_F$;
  Update error $\varepsilon_{\mathcal{C}} = \left\|\mathcal{C}^{s+1} - \mathcal{C}^s\right\|_F / \left\|\mathcal{C}^s\right\|_F$;
  if $\varepsilon_{\mathcal{X}} < tol$ and $\varepsilon_{\mathcal{C}} < tol$ **then**
    stop iteration.
  end if
end for

Output: image component $\mathcal{X}$ and cloud component $\mathcal{C}$.

low-rank prior knowledge. It is not necessary for the blind cloud removal method TVLRSDC to estimate cloud masks before recovering the clean images. The prior information that the blind method adopts includes smooth, sparse, and low-rank prior information. The difference between all the tested methods is summarized in Table I. From Table I, we can see that only BC-SLRpGS adopts the BC, which is proposed in this article and makes the method be a unified framework of cloud detection and removal. The experiments are implemented under Windows 10 and MATLAB version 9.3.0.713579 (R2017b) running on a desktop PC with an Intel Core i7-9700K CPU at 3.60 GHz and 16 GB of memory.

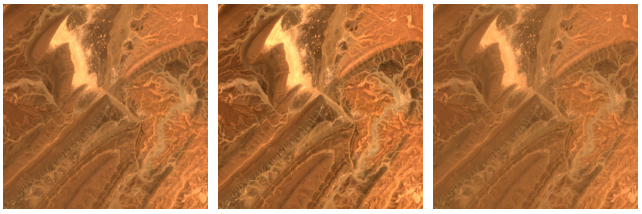
The mean peak signal-to-noise rate (PSNR), mean structural similarity index measure (SSIM) [72], and mean CC [73] values of reconstruction data $\mathcal{X} \in \mathbb{R}^{m\times n\times p\times q}$ are employed as the quantitative metrics, which are, respectively, defined by

$$\mathrm{PSNR} = \frac{1}{p|T|}\sum_{t\in T}\sum_{i=1}^{p} 10\log\frac{\left(\hat{\boldsymbol{X}}_i^t\right)_{\max}^2}{\left\|\boldsymbol{X}_i^t - \hat{\boldsymbol{X}}_i^t\right\|_F^2}$$

$$\mathrm{SSIM} = \frac{1}{p|T|}\sum_{t\in T}\sum_{i=1}^{p}\frac{\left(2\mu_{\boldsymbol{X}_i^t}\mu_{\hat{\boldsymbol{X}}_i^t} + c_1\right)\left(2\sigma_{\boldsymbol{X}_i^t\hat{\boldsymbol{X}}_i^t} + c_2\right)}{\left(\mu_{\boldsymbol{X}_i^t}^2 + \mu_{\hat{\boldsymbol{X}}_i^t}^2 + c_1\right)\left(\sigma_{\boldsymbol{X}_i^t}^2 + \sigma_{\hat{\boldsymbol{X}}_i^t}^2 + c_2\right)}$$

(a) (b) (c)

Fig. 3. Pesudocolor Brazil data taken on three different times by Sentinel-2. (a) June 1, 2020. (b) June 11, 2020. (c) June 21, 2020.



(a) (b) (c)

Fig. 4. Pesudocolor Morocco data taken on three different times by Sentinel-2. (a) June 25, 2020. (b) June 15, 2020. (c) July 5, 2020.

and

$$\mathrm{CC} = \frac{\sum_{j=1}^{k}(\boldsymbol{\mathcal{X}}_j - \mu_{\boldsymbol{\mathcal{X}}})(\hat{\boldsymbol{\mathcal{X}}}_j - \mu_{\hat{\boldsymbol{\mathcal{X}}}})}{\sqrt{\sum_{j=1}^{k}(\boldsymbol{\mathcal{X}}_j - \mu_{\boldsymbol{\mathcal{X}}})^2\sum_{j=1}^{k}(\hat{\boldsymbol{\mathcal{X}}}_j - \mu_{\hat{\boldsymbol{\mathcal{X}}}})^2}}$$

where T denotes the set of temporal nodes that are contaminated by clouds, $|T|$ denotes the cardinality of set T, $\hat{\boldsymbol{X}}_i^t$ and $\boldsymbol{X}_i^t$, respectively, denote the true image and reconstruction image of ith band and tth temporal node, $(\hat{\boldsymbol{X}}_i^t)_{\max}$ denotes the maximum pixel values of the image $\hat{\boldsymbol{X}}_i^t$; $\mu_{\boldsymbol{X}_i^t}$ and $\mu_{\hat{\boldsymbol{X}}_i^t}$, respectively, denote the mean values of $\boldsymbol{X}_i^t$ and $\hat{\boldsymbol{X}}_i^t$; $\sigma_{\boldsymbol{X}_i^t}$ and $\sigma_{\hat{\boldsymbol{X}}_i^t}$, respectively, denote the standard deviation of $\boldsymbol{X}_i^t$ and $\hat{\boldsymbol{X}}_i^t$; $\sigma_{\boldsymbol{X}_i^t\hat{\boldsymbol{X}}_i^t}$ denotes the covariance of $\boldsymbol{X}_i^t$ and $\hat{\boldsymbol{X}}_i^t$; c_1 and c_2 are two constants; $\boldsymbol{\mathcal{X}}_j$ and $\hat{\boldsymbol{\mathcal{X}}}_j$ are, respectively, the values of jth contaminated pixels for reconstruction and true images; $\mu_{\boldsymbol{\mathcal{X}}}$ and $\mu_{\hat{\boldsymbol{\mathcal{X}}}}$ are the corresponding mean values; and k denotes the number of contaminated pixels. The higher the three metrics values, the better the results.

A. Test Data

We study the performance of the proposed method with six synthetic datasets and three real datasets. The synthetic datasets are shown in Figs. 3–8. The data in Fig. 3 are taken over Brazil by Sentinel-2. We consider four bands at a ground

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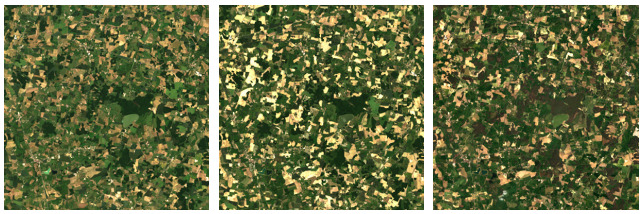
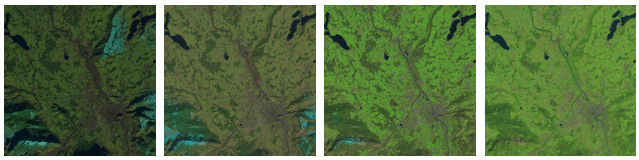


(a) (b) (c)

Fig. 5. Pesudocolor France data taken on three different times by Landsat-8. (a) June 13, 2019. (b) May 21, 2020. (c) April 3, 2020.

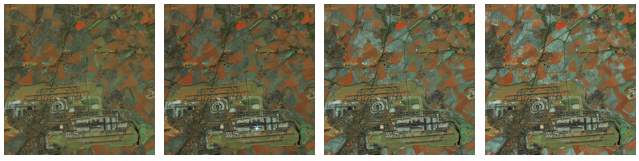
(a) (b) (c)

Fig. 6. Pesudo-color Picardie data (denoted as Picardie1) taken on three different times by Sentinel-2. (a) March 24, 2021. (b) March 24, 2021. (c) February 24, 2021.



(a) (b) (c) (d)

Fig. 7. Pesudocolor Munich data taken on four different times by Landsat-8. (a) January 2015. (b) March 2015. (c) April 2015. (d) August 2015.



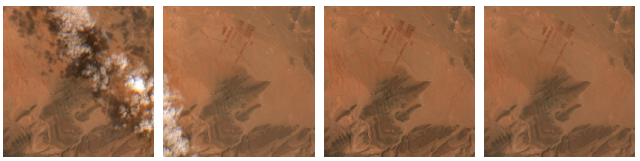
(a) (b) (c) (d)

Fig. 8. Pesudocolor Picardie data (denoted as Picardie2) taken on four different times by Sentinel-2. (a) March 24, 2021. (b) March 1, 2021. (c) February 24, 2021. (d) March 31, 2021.

sampling distance of 10 m (bands 2, 3, 4, and 8) taken on three different dates (June 1, 2020, June 11, 2020, and June 21, 2020). The data are of size $400 \times 400 \times 4 \times 3$. The data in Fig. 4 is taken over Morocco by Sentinel-2. We consider four bands at a ground sampling distance of 10 m (bands 2, 3, 4, and 8) taken on three different dates (June 25, 2020, June 15, 2020, and July 5, 2020). The data are of size $400 \times 400 \times 4 \times 3$. The data in Fig. 5 are taken over France by Landsat-8. We consider seven bands at a ground sampling distance of 30 m (bands 1–7) taken on three different dates (June 13, 2019, May 21, 2020, and April 3, 2020). The data are of size $400 \times 400 \times 7 \times 3$. The data in Fig. 7 are taken over Munich by Landsat-8. We consider three bands at a ground

(a) (b) (c) (d)

Fig. 9. Pesudocolor France data taken on four different times by Landsat-8. (a) February 24, 2020. (b) October 3, 2019. (c) February 8, 2020. (d) September 17, 2019.



(a) (b) (c) (d)

Fig. 10. Pesudocolor Morocco data taken on four different times by Sentinel-2. (a) August 22, 2020. (b) August 27, 2020. (c) August 17, 2020. (d) September 1, 2020.

(a) (b) (c) (d)

Fig. 11. Pesudocolor Eure data taken on four different times by Sentinel-2. (a) August 2017. (b) May 2017. (c) January 2017. (d) October, 2016.

sampling distance of 30 m (red, green, and blue) taken on four different dates (January 2015, March 2015, April 2015, and August 2015). The data are of size $512 \times 512 \times 3 \times 4$. The data in Figs. 6 and 8 are taken over Picardie, France, by Sentinel-2. We consider six bands at a ground sampling distance of 20 m (bands 5–7, 8A, 11, and 12) taken on three (March 24, 2021, March 1, 2021, and February 24, 2021) and four (March 24, 2021, March 1, 2021, February 24, 2021, and March 31, 2021) different dates, respectively. The data are of size $500 \times 500 \times 6 \times 3$ and $500 \times 500 \times 6 \times 4$, respectively.

The real datasets are shown in Figs. 9–11. The data in Fig. 9 are taken over France by Landsat-8. We consider seven bands at a ground sampling distance of 30 m (bands 1–7) taken on four different dates (February 24, 2020, October 3, 2019, February 8, 2020, and September 17, 2019). The data are of size $600 \times 600 \times 7 \times 4$. We can see that the data taken on February 24, 2020 and October 3, 2019, are contaminated by clouds. The data in Fig. 10 are taken over Morocco by Sentinel-2. Four bands at a ground sampling distance of 10 m (bands 2–4 and 8) taken on four different dates (August 22, 2020, August 27, 2020, August 17, 2020, and September 1, 2020) are considered. The data is of size $600 \times 600 \times 4 \times 4$. The data taken on August 22, 2020 and August 27, 2020 are covered by clouds. The data in Fig. 11 are taken over Eure by Sentinel-2. We consider four bands at a ground sampling distance of 10 m (bands 2–4 and 8) taken on four different dates

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Fig. 12. Recovered results for Brazil data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method.

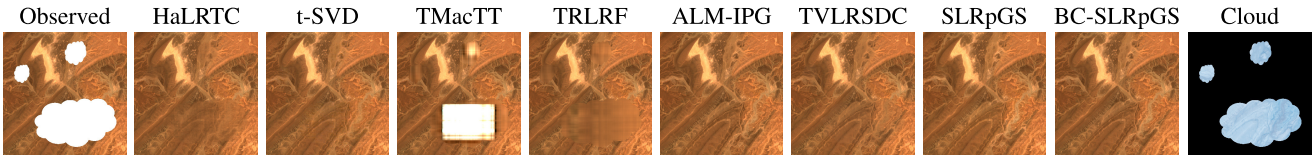

Fig. 13. Recovered results for Morocco data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method.

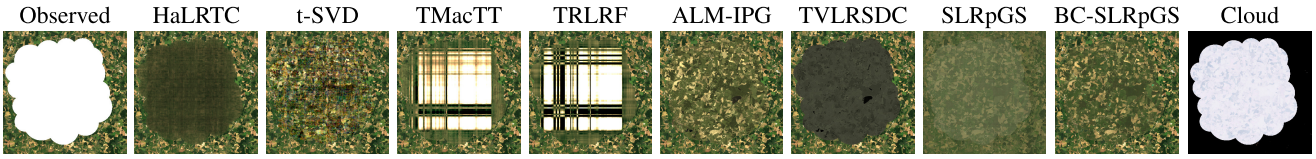

Fig. 14. Recovered results for France data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method.

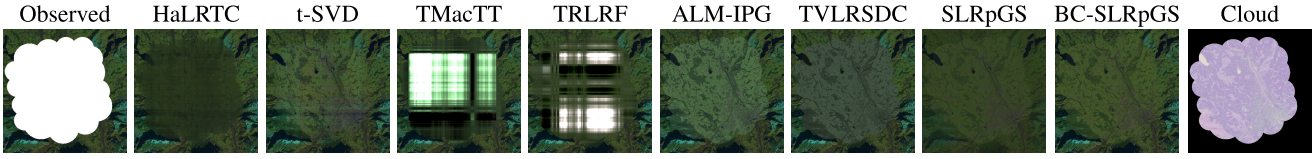

Fig. 15. Recovered results for Munich data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method.

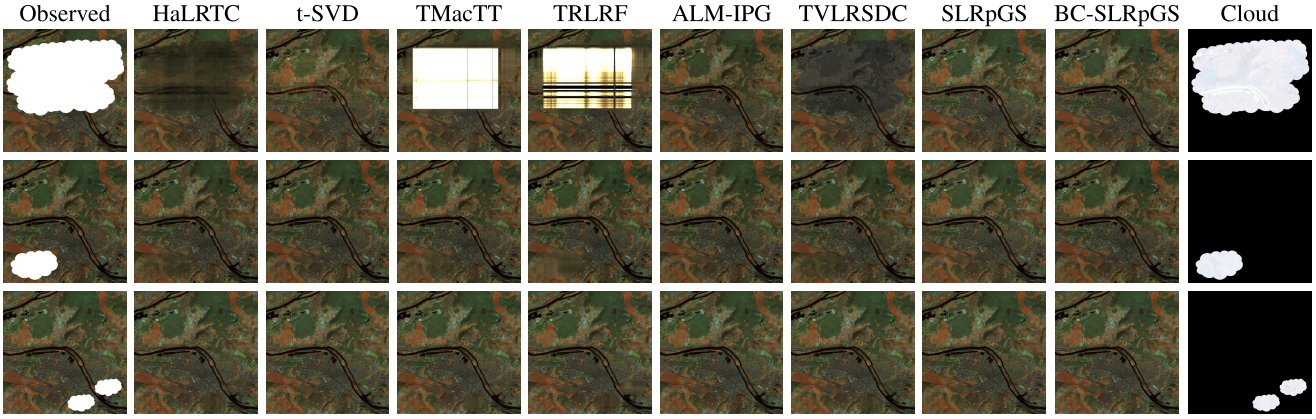

Fig. 16. Recovered results for Picardie1 data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method. From top to bottom: recovered results for temporal nodes 1–3, respectively.

(August 2017, May 2017, January 2017, and October 2016). The data are of size $400 \times 400 \times 4 \times 4$. We can see that the data taken in August 2017 are contaminated by clouds.

B. Synthetic Experiments

In this section, the simulated experiments are presented to test the proposed method BC-SLRpGS. The multitemporal data Brazil and Morocco are similar with respect to the temporal mode. We generate two simulated cloud masks, i.e., one is a single-large mask that is added to Brazil data and one is that there are three small masks that are added to Morocco data. The reconstructed results for Brazil and Morocco data are shown in Figs. 12 and 13, respectively. For Brazil results, t-SVD, AlM-IPG, SLRpGS, and BC-SLRpGS could obtain promising results. While the results of other methods, including the blind cloud removal method, TVLRSDC are visually worse. For Morocco data, except for HaLRTC, TMacTT, and TRLRF, other methods obtain promising results. From this comparison, it shows that the methods, HaLRTC, TMacTT,

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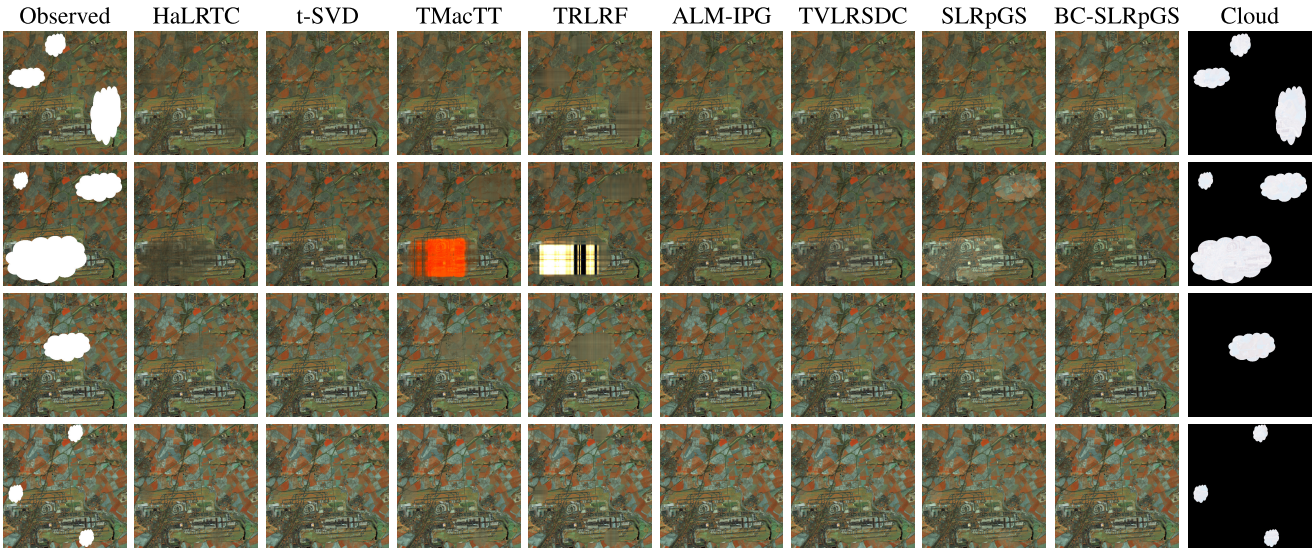

Fig. 17. Recovered results for Picardie2 data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method. From top to bottom: recovered results for temporal nodes 1–4, respectively.

TABLE II

PSNR AND SSIM RESULTS OF DIFFERENT METHODS FOR SIMULATED DATA. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Data	Metric	Observed	HaLRTC	t-SVD	TMacTT	TRLRF	ALM-IPG	TVLRSDC	SLRpGS	BC-SLRpGS
Brazil	PSNR	5.13	23.18	35.89	7.78	27.3	34.99	26.18	39.02	**42.51**
	SSIM	0.5324	0.8139	0.9633	0.5678	0.8052	0.9782	0.8768	0.9795	**0.9849**
	CC	0.0921	0.7683	0.9850	0.0160	0.8622	0.9847	0.8253	0.9945	**0.9960**
	Time (s)	0	12.3	50.8	278.5	179.2	19.0	146.1	306.3	360.3
Morocco	PSNR	8.93	39.22	44.55	12.33	36.16	50.86	43.98	42.68	**51.75**
	SSIM	0.7952	0.9730	0.9852	0.8308	0.9408	0.9964	0.9926	0.9901	**0.9967**
	CC	0.1275	0.9819	0.9944	0.1773	0.9620	0.9986	0.9933	0.9931	**0.9988**
	Time (s)	0	11.1	59.9	305.9	63.0	20.0	51.9	202.1	43.5
France	PSNR	4.15	26.67	27.51	9.53	7.7	28.27	25.00	27.03	**29.41**
	SSIM	0.4418	0.7317	0.7033	0.5377	0.4941	0.7748	0.7002	0.6674	**0.7944**
	CC	0.0304	0.5921	0.6355	-0.0411	-0.0004	0.6711	0.5276	0.4804	**0.7213**
	Time (s)	0	17.6	92.6	369.5	863.4	50.9	286.6	45.2	22.9
Picardie1	PSNR	10.01	45.45	52.95	36.47	34.13	54.51	46.29	47.59	**56.16**
	SSIM	0.8021	0.9774	0.9958	0.8776	0.8802	0.9982	0.9869	0.9939	**0.9985**
	CC	0.0881	0.8995	0.9851	0.6532	0.6788	0.9934	0.9395	0.9838	**0.9947**
	Time (s)	0	31.2	177.9	1024.5	4912.2	47.5	301.3	150.9	266.7
Munich	PSNR	4.29	25.78	26.66	6.45	8.73	23.26	26.23	25.53	**27.81**
	SSIM	0.4769	0.7172	0.8344	0.4993	0.5189	0.8462	0.8385	0.6985	**0.8506**
	CC	0.1479	0.7611	0.8353	0.1470	0.1641	0.7705	0.8026	0.7142	**0.8646**
	Time (s)	0	21.8	110.2	397.4	990.9	31.6	340.7	16.6	17.4
Picardie2	PSNR	10.58	47.71	52.16	44.88	40.04	54.08	52.65	44.81	**54.45**
	SSIM	0.8587	0.9911	0.9975	0.9712	0.9602	**0.9987**	0.9968	0.9902	0.9983
	CC	0.0737	0.9395	0.9734	0.8054	0.7180	**0.9775**	0.9716	0.9238	0.9756
	Time (s)	0	37.2	223.8	1235.5	2863.1	82.6	415.5	166.2	221.5

TRLRF, and TVLRSDC, are not good at handling the reconstruction task with large clouds. The proposed method could obtain promising results for both single/multiple small and large clouds removal.

We also test two simulated data, France and Munich data, whose temporal modes are not obviously similar. For the two data, the simulated cloud-contaminated data are generated by adding a large cloud mask. Figs. 14 and 15 show the recovered results by all compared methods, where we can see that the results obtained by t-SVD, ALM-IPG, and BC-SLRpGS are visually better than those of other methods. The result indicates that the methods except for t-SVD, ALM-IPG, and BC-SLRpGS could not recover promising results for the cloud-contaminated data whose temporal mode information is largely different.

What is more, we test two simulated cases where all the tested temporal modes are contaminated by clouds with the data Picardie. The results are shown in Figs. 16 and 17. From Figs. 16 and 17, we can see that the results for all temporal modes obtained by t-SVD, ALM-IPG, SLRpGS, and BC-SLRpGS are visually better compared with those of other methods. From Fig. 16, we can see that the blind method TVLRSDC cannot obtain promising results, especially for the first temporal mode that is contaminated by a large cloud.

The last columns of Figs. 12–17 are the absolute values of cloud component (i.e., $|\mathcal{C}|$) outputted by the proposed

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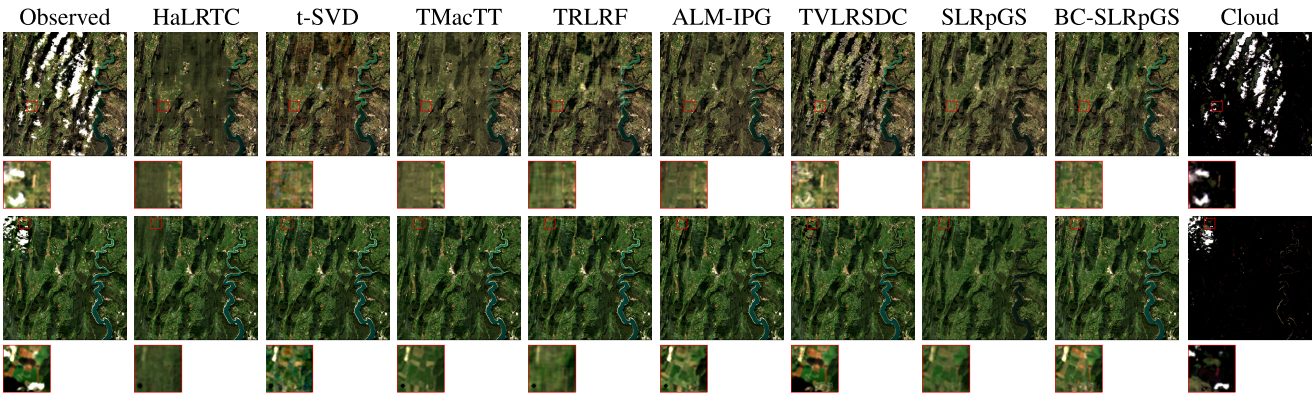

Fig. 18. Recovered results for France data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method. From top to bottom: recovered results for temporal nodes 1 and 2, respectively. The corresponding zoom-in patches are shown at the bottom of each image.

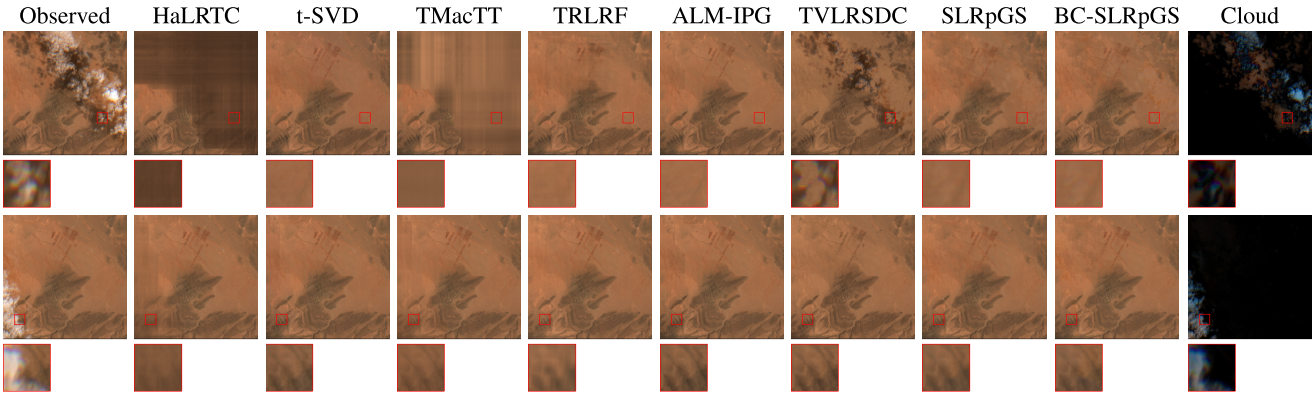

Fig. 19. Recovered results for Morocco data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method. From top to bottom: recovered results for temporal nodes 1 and 2, respectively. The corresponding zoom-in patches are shown at the bottom of each image.

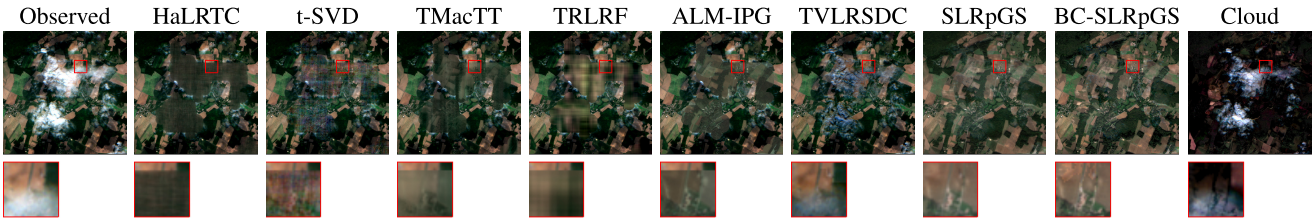

Fig. 20. Recovered results for Eure data by all compared methods. The last column is the cloud component obtained by BC-SLRpGS method. The corresponding zoom-in patches are shown at the bottom of each image.

method BC-SLRpGS, where we can see that they are almost the same with the simulated masks. The quantitative results of the above-mentioned simulated experiments are displayed in Table II, where we can see that the proposed method BC-SLRpGS obtains the best PSNR, SSIM, and CC results for Brazil, Morocco, France, and Picardie1 data. In addition, we have that the BC could improve the reconstructed results. In conclusion, the proposed method BC-SLRpGS could not only obtain competitive results but also generate accurate cloud components.

C. Real Experiments

In this section, we test three real data to study the performance of the proposed method. The first and second real data are taken over France by Landsat-8 and taken over Morocco by Sentinel-2, respectively. The first two temporal modes images of the two real data are contaminated by clouds. The reconstructed results are shown in Figs. 18 and 19. From Figs. 18 and 19, especially from the zoomed-in images, we can see that the results reconstructed by ALM-IPG, SLRpGS, and BC-SLRpGS are visually better than those by other methods. For the results of France, as shown in the first row of Fig. 18, the proposed method BC-SLRpGS could recover more smooth results and contains more details. For the results of Morocco, as shown in the first row of Fig. 19, all the methods except for HaLRTC, TMacTT, and TVLRSDC obtain promising results as this data taken at different times are similar. From the second rows of Figs. 18 and 19, it shows that the results reconstructed by all the methods except for HaLRTC are promising as the cloud-contaminated information is less compared with the cases ofk first rows.

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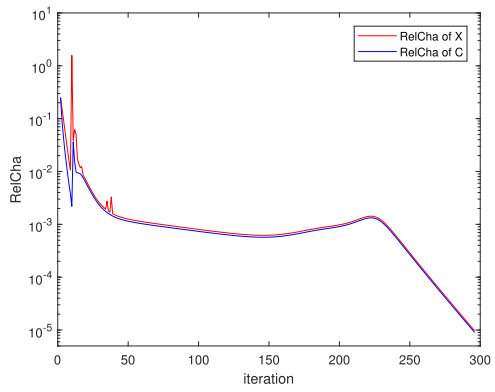

Fig. 21. RelCha of $\boldsymbol{\mathcal{X}}$ and $\boldsymbol{\mathcal{C}}$ with respect to iterations.

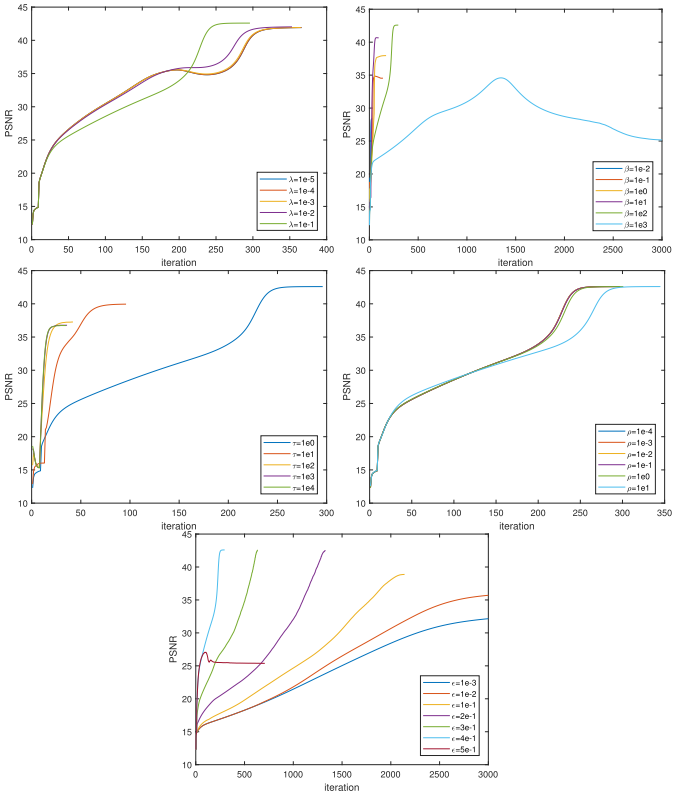

Fig. 22. PSNR values with respect to iterations under different parameters λ, $\beta = \beta_1 = \beta_2 = \beta_3$, τ, ρ, and ϵ.

The third real data are taken over Eure by Sentinel-2. One temporal mode image is contaminated by clouds. Fig. 20 shows the results of all compared methods. We can see that SLRpGS and BC-SLRpGS could reconstruct promising results, while other methods either could not reconstruct clear results for the area covered by clouds or could not detect the accurate cloud area. Besides, from Figs. [18]–[20], we can find that the method BC-SLRpGS could preserve more cloud-free information in the reconstructed results than the method without BC. It shows from the results of real data that the cloud component outputted by the proposed method contains some highlighted information that is not contaminated by clouds, this information actually is the shadow of clouds which are

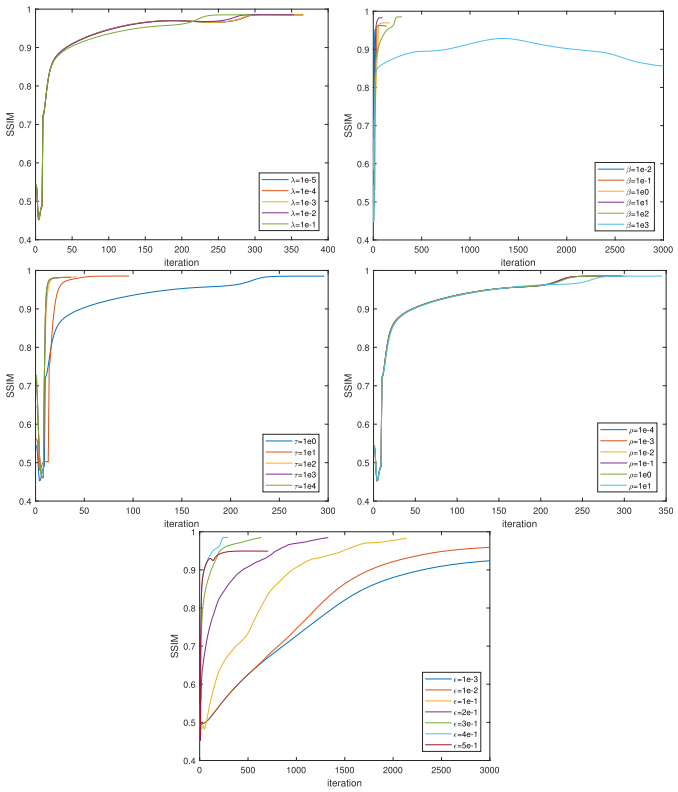

Fig. 23. SSIM values with respect to iterations under different parameters λ, $\beta = \beta_1 = \beta_2 = \beta_3$, τ, ρ, and ϵ.

also the noise. In conclusion, the reconstructed results by the proposed method are better than those of other methods quantitatively (PSNR, SSIM, and CC) and visually.

D. Discussions

In this section, we study the convergence analysis and the influence of parameters of the proposed method BC-SLRpGS with simulated Brazil data. Fig. 21 shows the relative change (RelCha) of the two successive estimated image component ($\boldsymbol{\mathcal{X}}$) and cloud component ($\boldsymbol{\mathcal{C}}$) with respect to iterations. Fig. 21 shows that the RelCha of $\boldsymbol{\mathcal{X}}$ and $\boldsymbol{\mathcal{C}}$ decreases globally as iterations, i.e., the convergence of the proposed method is numerically guaranteed. The PSNR and SSIM values with respect to iterations under different parameters λ, $\beta = \beta_1 = \beta_2 = \beta_3$, τ, ρ, and ϵ are shown in Figs. 22 and 23. From Figs. 22 and 23, we can see that the proposed method is robust to these parameters, especially for SSIM values. In Figs. 22 and 23, for the parameters β_1, β_2, and β_3, we study the case that they are set to be the same value β. For this case, the PSNR and SSIM values of proposed method are slightly different for different β. Furthermore, we separately study the PSNR and SSIM values under different values of β_1, β_2, and β_3, as shown in Fig. 24. Fig. 24 shows that the method could achieve the similar PSNR and SSIM values with different β_2 and β_3. For parameter β_1, the PSNR and SSIM values are nearly similar if β_1 is greater than 1. Overall, the proposed method is generally robust to the parameters

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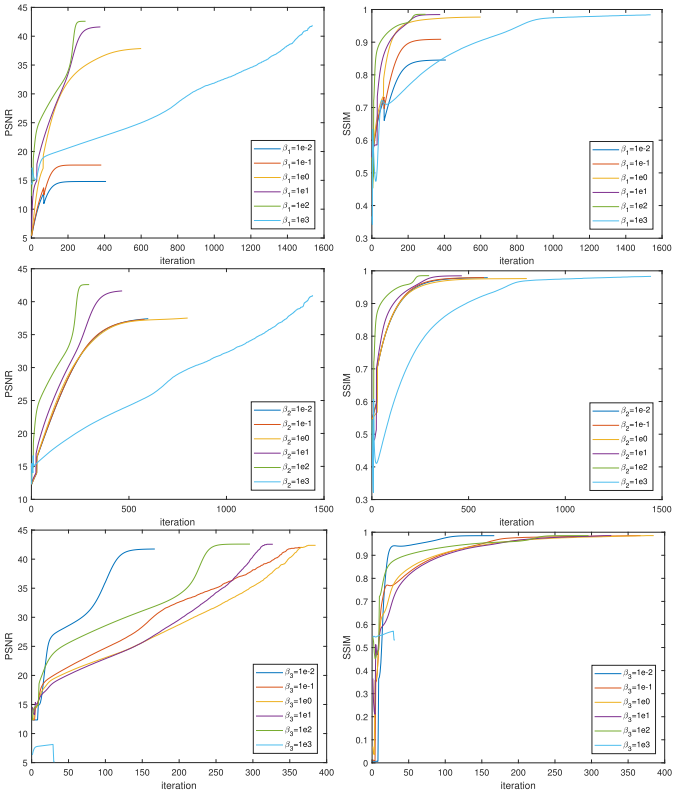

Fig. 24. PSNR and SSIM values with respect to iterations under different parameters β_1, β_2, and β_3.

and reaches the stop criterion with different iterations under different parameters.

IV. Conclusion

In this article, we proposed a group sparse function along the spectral dimension to characterize the cloud component inspired by the observation that a tube of the multispectral image contains cloud information if any pixel of this tube is contaminated by cloud. Based on the suggested function, we proposed a BC-SLRpGS model for cloud detection and removal. The proposed method does not need a cloud mask as prior knowledge or detect the mask before reconstructing the missing information, while the proposed method could estimate the missing information and detect the cloud mask simultaneously. In addition, the proposed model integrates a BC, which can enforce the cloud-free pixel values of recovered data to be equal to those of observed data in each iterative step. An efficient algorithm that alternately generates the cloud mask, image component, and cloud component was developed. Extensive experiments on synthetic and real data show that the proposed method is competitive compared with the tensor completion and blind cloud removal methods.

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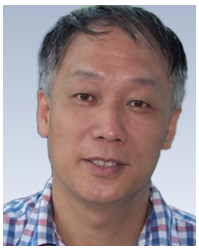

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